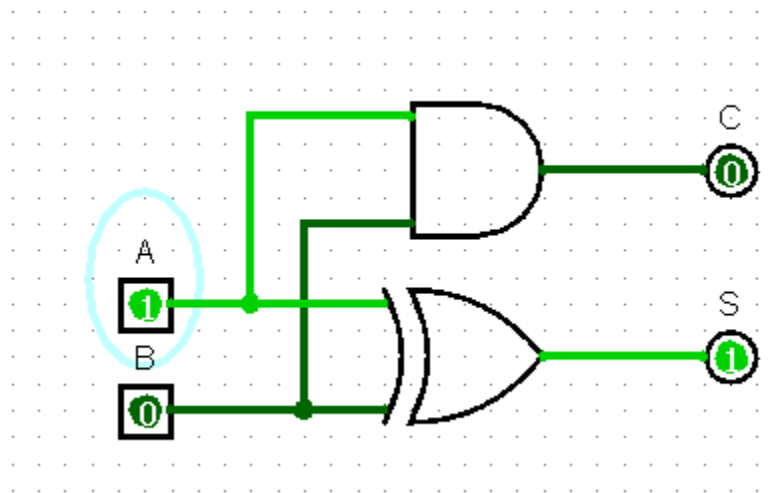


**USTP IC DESIGN AND VERIFICATION**  
**DIGITAL LOGIC DESIGN**  
**WEEK 2**

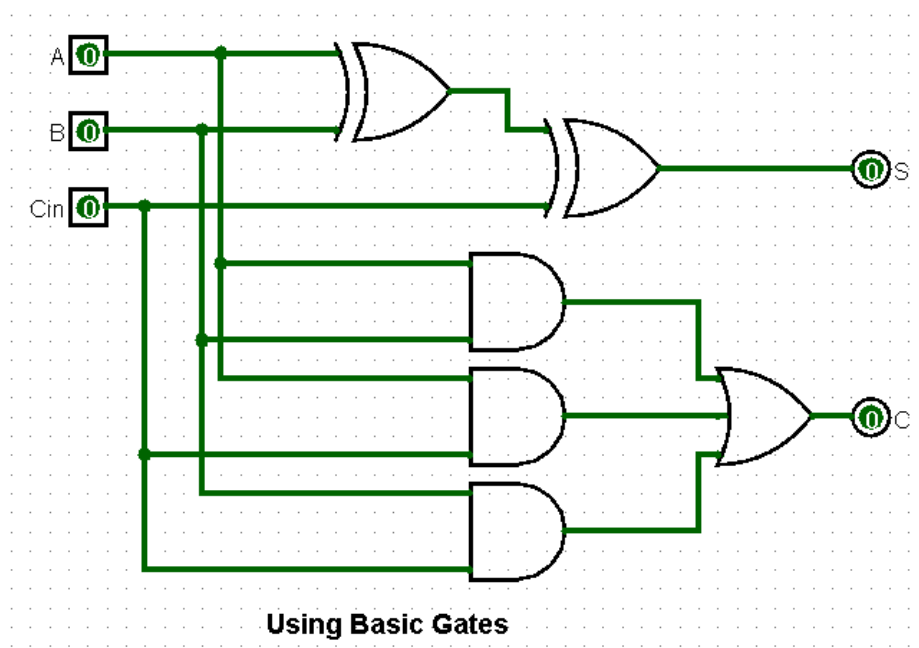
**NAME: Ali Irfan**

**ROLL NO: 22FA-036-CE**

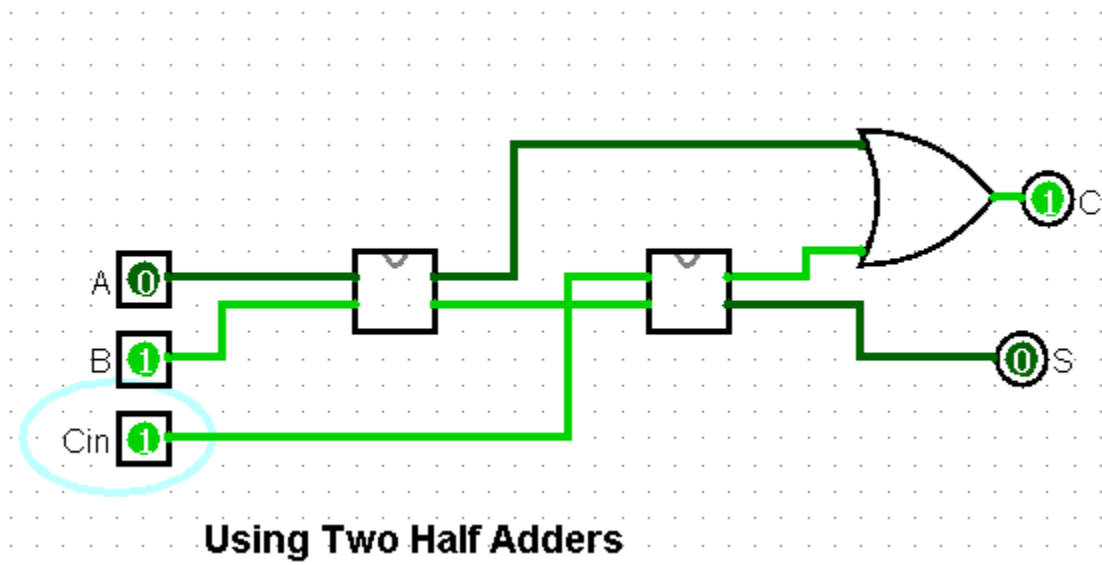
**1) HALF ADDER**



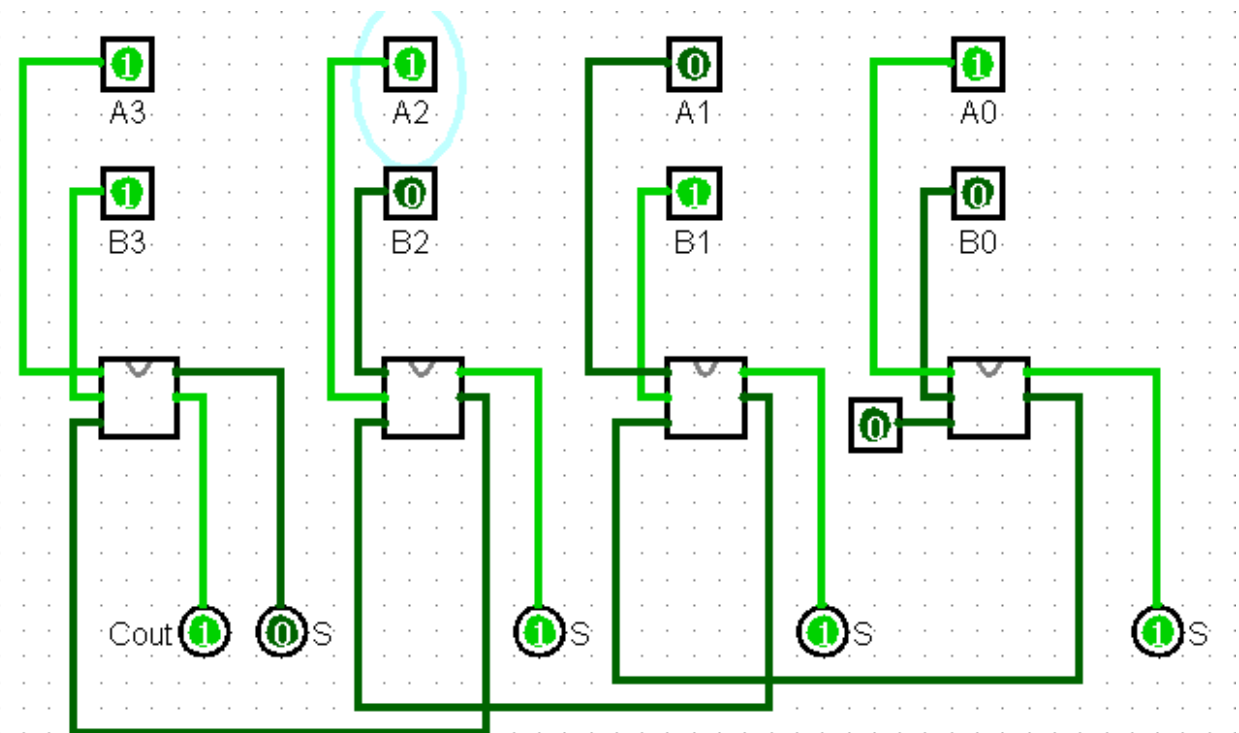
**2) FULL ADDER USING BASIC GATES**



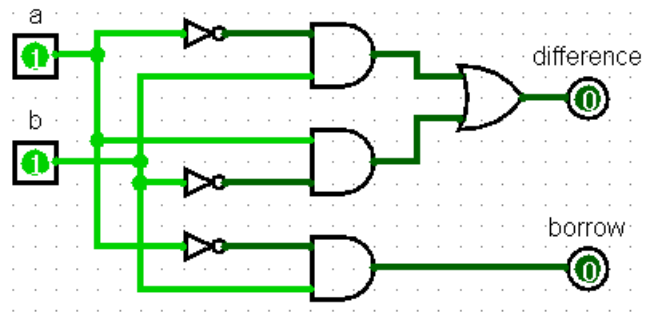
### 3) FULL ADDER USING 2 HALF ADDERS



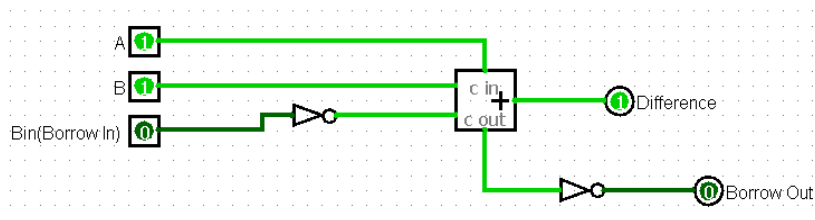
### 4) 4 BIT RIPPLE CARRY ADDER



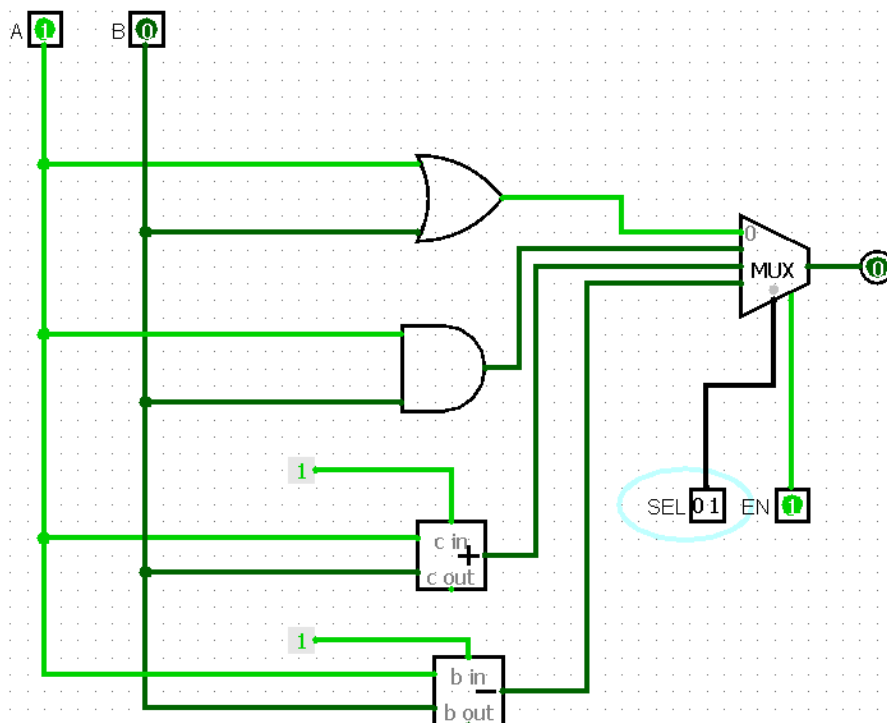
## 5) HALF SUBTRACTOR



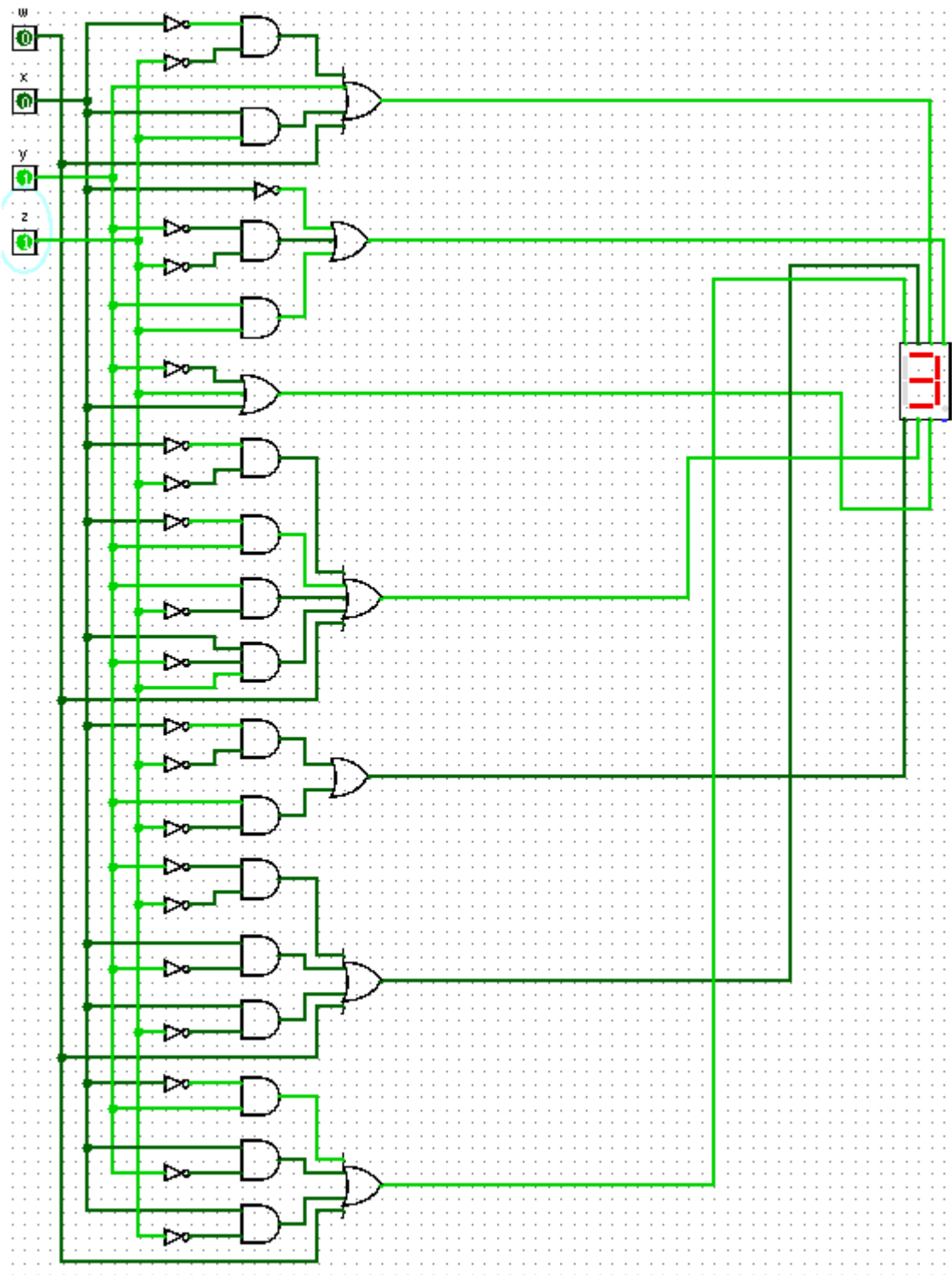
## 6) FULL SUBTRACTOR



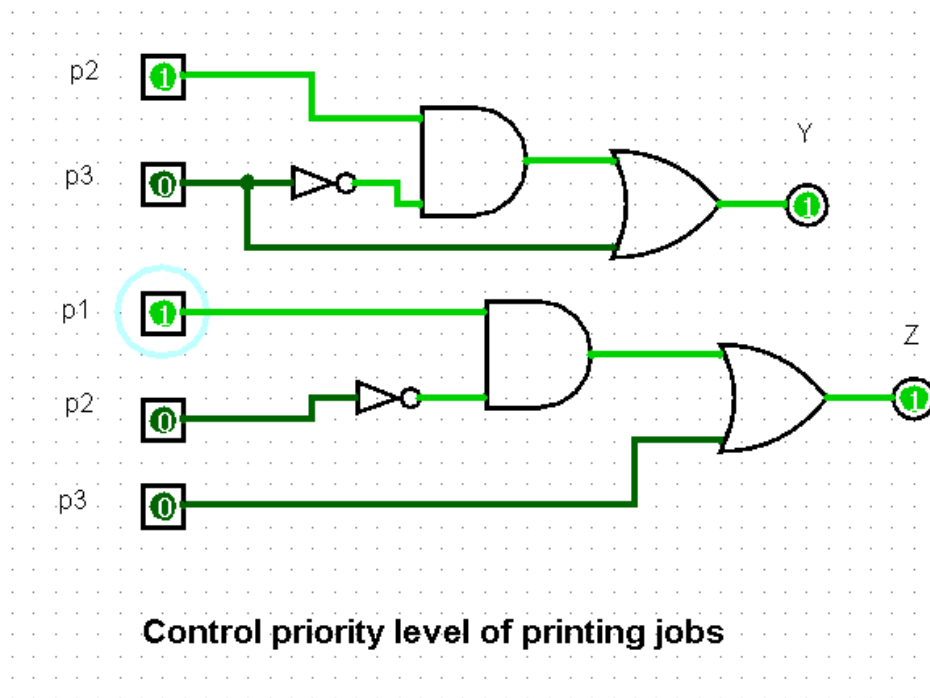
## 7) ALU



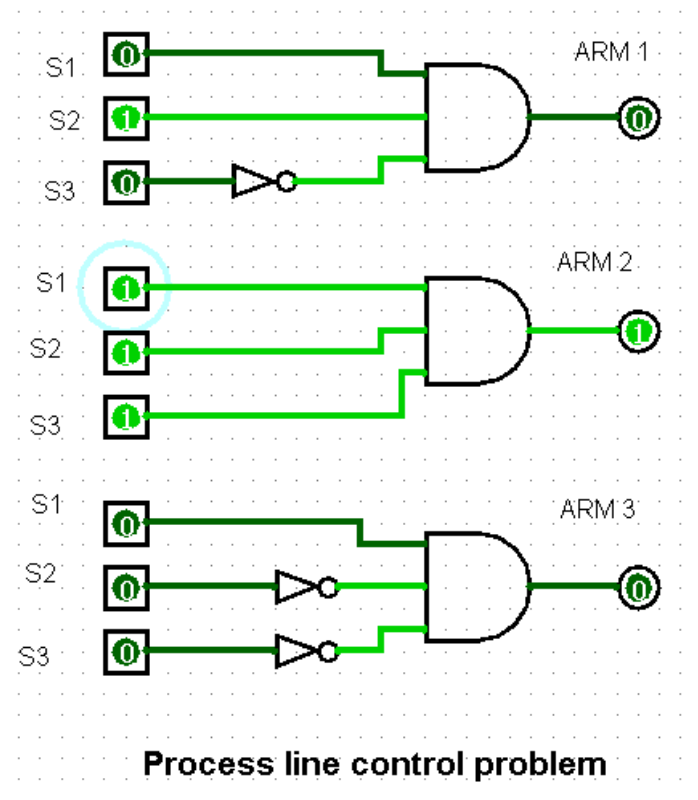
## 8) BCD TO 7 SEGMENT



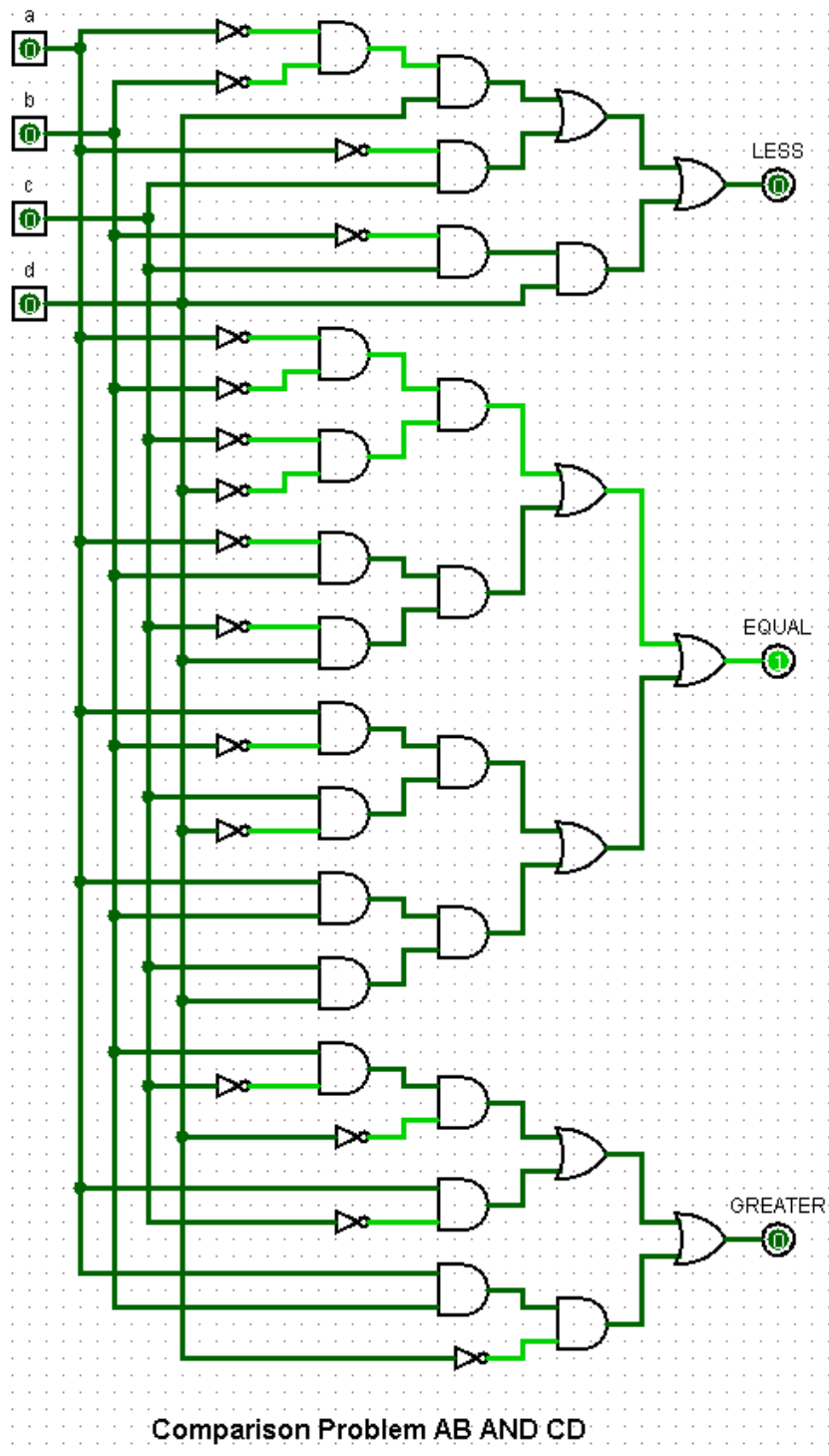
## 9) CHALLENGE ACTIVITY 1



## 10) CHALLENGE ACTIVITY 2



### 11) CHALLENGE ACTIVITY 3



## 12) CHALLENGE ACTIVITY 4

